
BRIDGING THE REALITY GAP: DUAL OPTIMAL TRANSPORT POLICY GRADIENT FOR REAL-WORLD REINFORCEMENT LEARNING FROM DEMONSTRATIONS

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ABSTRACT

A fundamental obstacle to deploying reinforcement learning (RL) in real-world settings is the pervasive reality gap—a systematic discrepancy that causes policies trained in simulation to fail when deployed on physical systems. Although learning from demonstrations (LfD) provides a promising alternative to manual reward engineering, current approaches often depend on imperfect predefined reward functions or encounter issues with distribution shift, which limits their practical applicability. To address these challenges, this paper introduces the Dual Optimal Transport Policy Gradient (DOT-PG) framework, which uses optimal transport duality to learn robust policies directly from expert trajectories. By minimizing the Wasserstein distance between the state-action distributions of the agent and the expert, DOT-PG automatically extracts both a stable policy gradient and a robust value function from the optimal transport dual variables—entirely removing the need for handcrafted reward design. The proposed method offers theoretical guarantees for policy improvement and is designed to learn policies that remain invariant under sim-to-real dynamics mismatches. Extensive experiments show that DOT-PG substantially outperforms state-of-the-art imitation learning and RLfD baselines in zero-shot sim-to-real transfer success rate, sample efficiency, and asymptotic performance, establishing a new paradigm for training real-world policies from demonstration data.

Keywords Sim-to-Real Transfer, Optimal Transport, Policy Gradient, Reinforcement Learning from Demonstrations, Robust Reinforcement Learning

1 Introduction

The paradigm of reinforcement learning (RL) has demonstrated remarkable unprecedented success across diverse domains, from mastering complex games to enabling sophisticated robotic behaviors. However, the transition of RL methodologies from controlled simulated environments to real-world physical systems remains fraught with fundamental persistent challenges. Central among these is the “reality gap”—the systematic inherent discrepancy between simulation and reality that manifests through unmodeled dynamics, sensor inaccuracies, actuator delays, and environmental uncertainties. This gap represents a critical fundamental bottleneck that prevents the widespread deployment of RL in safety-critical applications such as autonomous driving, medical robotics, and industrial automation.

The challenge of the reality gap is particularly acute in domains requiring fine-grained precise physical interactions. While sophisticated high-fidelity simulators like MuJoCo, Drake and Issac Sim have advanced significantly, they inevitably incorporate certain deliberate simplifications and approximations that diverge from physical reality. These discrepancies, though often imperceptible during training, can lead to sudden catastrophic failures during deployment. A policy that appears optimal in simulation may fail abruptly and completely when faced with subtle real-world friction, sensor noise, or unmodeled dynamics. This pressing challenge has spurred urgent significant research interest in developing algorithms that can learn robust policies capable of bridging the simulation-to-reality divide.

Learning from demonstrations (LfD) has emerged as a promising approach to address this challenge. By leveraging expert trajectories, LfD methods seek to bypass the need for meticulous reward engineering—a process that is particularly challenging in real-world scenarios where reward signals may be sparse, poorly shaped, or difficult to specify. However, current LfD methodologies face their own fundamental limitations, such as compounding causal distribution shift, inherent sensitivity to noisy demonstrations, over-reliance on prohibitively expensive large datasets, and poor out-of-distribution generalization capabilities, when applied to real-world settings.

Behavioral cloning (BC), while conceptually simple and straightforward to implement, suffers from the well-documented problem of compounding errors. The supervised learning framework of BC fails to account for the sequential nature of decision-making, leading to cascading failures when the agent encounters states outside the expert’s distribution. Techniques like Dataset Aggregation (DAgger) partially mitigate this issue through interactive expert querying, but this approach requires substantial human involvement and may be impractical for many real-world applications.

Inverse reinforcement learning (IRL) and its modern variants offer a more principled alternative by seeking to recover the underlying reward function that explains expert behavior. However, IRL methods typically involve solving a nested optimization problem that requires repeated policy evaluation in an inner loop, making them computationally prohibitive for complex tasks. Moreover, the reward functions recovered by IRL may not generalize well across the reality gap, as they are often tied to features that are specific to the simulation environment.

The advent of adversarial imitation learning, particularly Generative Adversarial Imitation Learning (GAIL), represented a significant advancement by framing imitation as a distribution matching problem. While GAIL and its variants have achieved impressive results, they inherit the training instabilities characteristic of generative adversarial networks. These methods are often sensitive to hyperparameter choices, prone to mode collapse, and may produce policies that exploit simulator-specific artifacts rather than learning robust behaviors.

The fundamental limitation shared by these approaches is their reliance on metrics that may not adequately capture the geometric structure of the policy space. f -divergences, which underlie many adversarial methods, can provide misleading signals when distributions have limited support overlap—a common scenario in real-world applications where data is scarce. This limitation becomes particularly problematic when policies must generalize across the reality gap, where the training and deployment distributions may have significant divergence.

On the other hand, optimal transport (OT) theory offers a principled rigorous framework that directly addresses these core limitations. The Wasserstein distance, unlike typical f -divergences, provides a smooth and meaningful geometric measure of distribution discrepancy even when distributions have completely disjoint supports. This crucial property makes OT particularly well-suited for imitation learning scenarios where limited expert demonstrations may cover only a small portion of the relevant state-action space. More importantly, the dual formulation of optimal transport provides a natural and direct connection to value function and generalized advantage learning in reinforcement learning.

The Kantorovich-Rubinstein duality establishes that the Wasserstein distance can be expressed as a supremum over Lipschitz functions. This dual perspective reveals that the optimal critic in the OT formulation serves simultaneously as a measure of distribution discrepancy and as a potential-based reward signal for policy improvement. This insight forms the theoretical foundation of our approach, bridging the gap between distribution matching and policy optimization.

In this paper, we introduce the novel Dual Optimal Transport Policy Gradient (DOT-PG) framework, which systematically leverages optimal transport duality to enable provably robust reinforcement learning directly from expert demonstrations trajectories. Our principled approach is grounded in several fundamental key insights:

First, by formulating imitation learning as a Wasserstein distance minimization problem, we obtain a smooth and well-behaved optimization landscape that facilitates stable training and better convergence properties. The Lipschitz continuity requirements inherent in the OT dual formulation naturally enforce regularization on the learned value functions, preventing the explosive gradients that often plague adversarial methods.

Second, the optimal dual variables in our framework serve multiple purposes, that is, they not only provide a dense learning signal for policy improvement, but also act as a robust value function estimate, and at the same time naturally yield an advantage function for policy gradients. This multi-functional role eliminates the need for separate reward learning and policy optimization phases, leading to greater algorithmic simplicity and efficiency.

Third, the geometric properties of the Wasserstein distance make our approach particularly suitable for real-world applications. By focusing on the underlying metric structure of the state-action space rather than the probability density ratios, the proposed DOT-PG algorithm learns policies that are more robust to distribution shifts and dynamics mismatches, which usually lead to the challenges in dealing with the sim-to-reality gap.

The contributions of this work are fourfold, encompassing both theoretical foundations and practical implementations:

First, we propose DOT-PG, a novel framework that for the first time unifies optimal transport duality with policy gradient methods, establishing a new paradigm for reinforcement learning directly from demonstrations.

Second, we establish comprehensive theoretical guarantees demonstrating that our approach ensures monotonic policy improvement through bounded policy updates with provable convergence. The derived policy gradients exhibit inherent robustness properties fundamentally due to the Lipschitz continuity enforced by the OT dual formulation, providing rigorous mathematical foundations for stable and convergent learning under well-behaved state distributions.

Third, we develop practical, efficient, and scalable algorithmic implementations that maintain theoretically crucial Lipschitz boundedness through differentiable normalization and learnable adaptive constraint mechanisms. This design ensures certified convergence properties while enabling robustness against dynamics mismatch, observation noise, unmodeled effects and environmental variability, effectively avoiding complex density ratio estimation.

Fourth, we conduct extensive experimental validation demonstrating DOT-PG’s performance robustness and policy convergence in challenging sim-to-real transfer. In general, our method outperforms state-of-the-art baselines while maintaining persistent stable performance and consistent convergence behavior across diverse continuous control tasks.

The DOT-PG framework represents a significant step toward making reinforcement learning practically viable for real-world systems. By leveraging the mathematical foundations of optimal transport theory, we provide a principled solution to the challenge of learning robust policies from demonstrations that can successfully bridge the reality gap.

2 Related Work

2.1 Imitation Learning Paradigms

Imitation learning has evolved through several distinct methodological approaches, each with unique strengths and limitations in addressing the reality gap challenge. Behavioral Cloning (BC) represents the most straightforward approach, framing imitation as a supervised learning problem over expert state-action pairs. While computationally efficient, BC suffers from fundamental limitations including compounding errors and covariate shift, where small errors accumulate during execution leading to progressively worsening performance. Pioneering work by Ross et al. (2011) introduced Dataset Aggregation (DAgger) to address these issues through iterative data collection and expert relabeling, though this approach requires ongoing expert involvement that may be impractical for real-world deployment.

Inverse Reinforcement Learning (IRL) approaches, initiated by Ng and Russell (2000), take a fundamentally different perspective by attempting to recover the underlying reward function that explains expert behavior. The Maximum Entropy IRL framework (Ziebart et al., 2008) advanced the field by providing a probabilistic formulation that resolves reward ambiguity, while later extensions like Guided Cost Learning (Finn et al., 2016) and Adversarial Inverse Reinforcement Learning (AIRL) (Fu et al., 2018) improved scalability through deep learning integration. However, these methods typically require solving a nested reinforcement learning problem in an inner loop, making them computationally intensive and challenging to tune for complex tasks.

2.2 Adversarial Imitation Learning

The introduction of Generative Adversarial Imitation Learning (GAIL) (Ho & Ermon, 2016) marked a significant paradigm shift by framing imitation as a distribution matching problem using adversarial training. GAIL employs a discriminator to distinguish between expert and agent trajectories while the policy learns to generate behavior that fools this discriminator. Subsequent improvements include InfoGAIL (Li et al., 2017), which incorporates latent codes for skill discovery, and VAIL (Peng et al., 2019), which adds information-theoretic constraints for improved sample efficiency. Despite their empirical success, these methods inherit the training instabilities characteristic of generative adversarial networks, including mode collapse, sensitivity to hyperparameters, and difficulty in achieving convergence—issues that become particularly problematic in real-world applications where training stability is crucial.

2.3 Optimal Transport in Machine Learning

Optimal transport theory has emerged as a powerful framework for comparing probability distributions in machine learning. The Wasserstein GAN (Arjovsky et al., 2017) demonstrated the practical benefits of using Wasserstein distances for generative modeling, providing improved training stability and meaningful loss metrics. In domain adaptation, Wasserstein Distance Guided Representation Learning (Shen et al., 2018) showed that OT metrics can effectively align distributions across domains. More recently, optimal transport has been applied to reinforcement learning through methods like Wasserstein Adversarial Imitation Learning (WAIL) (Xiao et al., 2019) and Wasserstein

Inverse Reinforcement Learning (WIRL) (Ni et al., 2020), which replace f-divergences with Wasserstein distances but often maintain the adversarial training framework with its associated instability.

2.4 Policy Gradient Methods

Policy gradient methods form the foundation of modern deep reinforcement learning in continuous action spaces. The Deep Deterministic Policy Gradient (DDPG) (Lillicrap et al., 2015) extended the deterministic policy gradient theorem to deep neural networks, while Twin Delayed DDPG (TD3) (Fujimoto et al., 2018) addressed overestimation bias through clipped double Q-learning and delayed policy updates. The Soft Actor-Critic (SAC) (Haarnoja et al., 2018) framework incorporated maximum entropy reinforcement learning for improved exploration and robustness. While these methods have achieved state-of-the-art performance in simulated environments, they typically require carefully engineered reward functions and struggle with the sim-to-real transfer problem due to their sensitivity to reward specification and dynamics mismatch.

2.5 Bridging the Reality Gap

Several approaches have been developed specifically to address the simulation-to-reality transfer challenge. Domain Randomization (Tobin et al., 2017) exposes policies to varied simulated conditions to encourage robustness, while System Identification (Yu et al., 2017) attempts to precisely match simulation parameters to real-world dynamics. Meta-learning approaches (Finn et al., 2017) aim to learn policies that can quickly adapt to new environments, and Robust Reinforcement Learning methods (Morimoto & Doya, 2005; Pinto et al., 2017) explicitly optimize for worst-case performance under dynamics uncertainty. However, these approaches often require extensive real-world data collection or make conservative assumptions that limit asymptotic performance.

2.6 Integration with Imitation Learning

Recent work has begun to explore the intersection of imitation learning and sim-to-real transfer. SimOpt (Ramos et al., 2019) combines domain randomization with demonstration-guided policy search, while DART (James et al., 2019) uses adversarial discriminators to align simulation and real-world data distributions. However, these methods typically maintain the adversarial training framework with its associated instabilities and high sample complexity, or require complex multi-stage optimization procedures that are sensitive to hyperparameter tuning.

2.7 Highlights of DOT-PG Algorithm

Compared with these results, our proposed Dual Optimal Transport Policy Gradient (DOT-PG) framework distinguishes itself from prior work by directly leveraging optimal transport duality to connect distribution matching with policy optimization in a single coherent framework. Unlike adversarial methods, DOT-PG provides theoretical guarantees of policy improvement and training stability through its Lipschitz-constrained optimization. Unlike traditional IRL approaches, it avoids the computationally expensive inner reinforcement learning loop. And unlike prior OT-based imitation learning methods, it fully exploits the connection between OT dual variables and value functions to enable direct policy gradient computation without intermediate reward learning. This integrated approach allows DOT-PG to simultaneously address the challenges of sample efficiency, training stability, and sim-to-real robustness that have limited previous methods, providing a principled foundation for real-world reinforcement learning from demonstrations.

3 Theoretical Foundations

3.1 Optimal Transport Theory

Optimal transport theory provides a powerful geometric framework for comparing probability distributions by considering the minimal cost required to transform one distribution into another. The theory has evolved through two fundamental formulations that provide complementary perspectives on distribution matching.

Monge Formulation: The classical Monge formulation seeks a deterministic transport map that minimizes the transportation cost. Precisely, given two probability measures μ and ν defined on metric spaces $\mathcal{X}$ and $\mathcal{Y}$ respectively, and a cost function $c : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^+$, the Monge problem aims to find a transport map $T : \mathcal{X} \rightarrow \mathcal{Y}$ that minimizes

$$\inf_{T \# \mu = \nu} \int_{\mathcal{X}} c(x, T(x)) d\mu(x) \quad (1)$$

where $T_\# \mu$ denotes the pushforward measure of μ under T , requiring that T transports the mass from μ to ν . While conceptually elegant, this formulation is often too restrictive as it requires the existence of a deterministic transport map and may not have a solution when μ contains atoms but ν does not.

Kantorovich Relaxation: The Kantorovich formulation generalizes the Monge problem by considering probabilistic couplings rather than deterministic maps. This relaxation admits solutions under much more general conditions, i.e.,

$$W_c(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} c(x, y) d\gamma(x, y) \quad (2)$$

where $\Pi(\mu, \nu)$ denotes the set of all joint distributions (couplings) with marginals μ and ν . The coupling γ specifies how mass is transported from μ to ν , allowing for splitting mass from a single source point to multiple target points.

Wasserstein Distance: For $p \geq 1$, the p -Wasserstein distance provides a family of metrics on probability distributions

$$W_p(\mu, \nu) = \left(\inf_{\gamma \in \Pi(\mu, \nu)} \int_{\mathcal{X} \times \mathcal{Y}} d(x, y)^p d\gamma(x, y) \right)^{1/p} \quad (3)$$

where $d : \mathcal{X} \times \mathcal{Y} \rightarrow \mathbb{R}^+$ is a metric on the underlying space. It should be emphasized that the case $p = 1$ is particularly important due to its connection to the Kantorovich-Rubinstein duality which will be discussed in Theorem 1.

Connection to Our Result: We consider the state-action distributions ρ_π (agent) and ρ_E (expert) defined on the joint state-action space $\mathcal{S} \times \mathcal{A}$. The key advantage of Wasserstein distance over f-divergences (such as KL-divergence or JS-divergence) is its ability to provide meaningful gradients even when distributions have non-overlapping support. This property makes it particularly suitable for imitation learning, where the agent's initial distribution may be far from the expert distribution, and ensures smooth and stable training dynamics throughout the learning process.

What's more, the geometric nature of optimal transport allows it to capture the underlying structure of the state-action space, making it robust to variations in policy parameterization and environmental dynamics, as it enables policies to maintain expert-like behavior even under distribution shifts between simulation and real-world deployment.

Essentially, the optimal Transport theory provides a geometric framework for comparing probability distributions by finding the minimal cost required to transform one distribution into another. Given two probability distributions ρ_π (agent) and ρ_E (expert) defined on a metric space $\mathcal{X}$, and for any $p \geq 1$, the p -Wasserstein distance is defined as below

$$W_p(\rho_\pi, \rho_E) = \left(\inf_{\gamma \in \Pi(\rho_\pi, \rho_E)} \int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p d\gamma(x, y) \right)^{1/p}$$

where $\Pi(\rho_\pi, \rho_E)$ denotes the set of all joint distributions (couplings) with marginals ρ_π and ρ_E , $d(x, y)$ is a metric on $\mathcal{X}$, typically the Euclidean distance, γ represents the transport plan that specifies how mass is moved from ρ_π to ρ_E .

3.2 Kantorovich Duality and Entropic Regularization

The Kantorovich duality and entropic regularization provide the mathematical foundation that connects primal optimal transport problems with their dual formulations, and enable efficient computation and theoretical analysis.

3.2.1 Fundamental Duality Theorems

The fundamental theorem of Kantorovich duality establishes the dual formulation for the 1-Wasserstein distance.

Theorem 1 (Kantorovich-Rubinstein Duality). *For the 1-Wasserstein distance with metric cost $d(x, y)$, we have*

$$W_1(\mu, \nu) = \sup_{f \in \text{Lip}_1} \left[\int_{\mathcal{X}} f(x) d\mu(x) - \int_{\mathcal{Y}} f(y) d\nu(y) \right] \quad (4)$$

where Lip_1 is the set of 1-Lipschitz functions $|f(x_1) - f(x_2)| \leq d(x_1, x_2)$, $d(x_1, x_2)$ is the distance between x_1, x_2 .

Proof. Based on the Fenchel-Rockafellar duality framework, let us consider the following two convex functionals

$$\Phi(f) = \int f d\mu - \int f d\nu, \quad \Psi(f) = \begin{cases} 0 & \text{if } f \in \text{Lip}_1 \\ +\infty & \text{otherwise} \end{cases} \quad (5)$$

Then, we give the conjugate functions of $\Phi(f)$ and $\Psi(f)$, which is calculated by the supremums of integral functions

$$\Phi^*(g) = \sup_f \left\{ \int f g - \Phi(f) \right\} = \sup_f \left\{ \int f(g + \nu - \mu) \right\}, \quad \Psi^*(g) = \sup_{f \in \text{Lip}_1} \int f g. \quad (6)$$

Note that $\Phi^*(g)$ is finite only when $g = \mu - \nu$ in the distributional sense, otherwise it diverges to $+\infty$. Obviously, this is because the supremum over f of $\int f(g + \nu - \mu)$ is unbounded unless the equation $g + \nu - \mu = 0$ is imposed.

Next, by the Fenchel-Rockafellar duality theorem, it can be directly derived that the following equation holds

$$\sup_f [\Phi(f) - \Psi(f)] = \inf_g [\Psi^*(g) + \Phi^*(-g)] \quad (7)$$

It can be seen that the left-hand side of this equation is exactly our primal problem of Fenchel-Rockafellar duality

$$\sup_{f \in \text{Lip}_1} \left[\int f d\mu - \int f d\nu \right] \quad (8)$$

For the right-hand side of this equation, substituting our conjugate functions gives the following expression

$$\inf_g [\Psi^*(g) + \Phi^*(-g)] = \inf_g \left[\sup_{f \in \text{Lip}_1} \int f g + \Phi^*(-g) \right] \quad (9)$$

Since $\Phi^*(-g)$ implicitly imposed the constraints $-g = \nu - \mu$ (or equivalently $g = \mu - \nu$), thus it implies

$$\inf_{g=\mu-\nu} \sup_{f \in \text{Lip}_1} \int f g = \sup_{f \in \text{Lip}_1} \int f d(\mu - \nu) \quad (10)$$

This confirms the Fenchel-Rockafellar duality and yields the desired 1-Wasserstein distance formulation, that is

$$W_1(\mu, \nu) = \sup_{f \in \text{Lip}_1} \left[\int f d\mu - \int f d\nu \right]. \quad \square$$

The conclusion can be extended for general cost functions, i.e., we have the following more general Kantorovich duality.

Theorem 2 (General Kantorovich Duality). *For any lower semi-continuous cost function $c(x, y)$, we have the equation*

$$\inf_{\gamma \in \Pi(\mu, \nu)} \int c d\gamma = \sup_{(f, g) \in \Phi_c} \left[\int f d\mu + \int g d\nu \right] \quad (11)$$

where Φ_c is defined by the set of (f, g) , i.e., $\Phi_c = \{(f, g) \in L^1(\mu) \times L^1(\nu) : f(x) + g(y) \leq c(x, y) \text{ for all } x, y\}$.

All these duality results hold particular significance for our proposed DOT-PG framework, as they fundamentally transform the computationally challenging optimization over transport couplings into a more tractable functional optimization problem. This reformulation enables the optimal dual variables f^* and g^* to be naturally interpreted as meaningful quantities that can be effectively approximated using neural networks. Moreover, in the 1-Wasserstein formulation, the inherent Lipschitz constraint serves as a powerful regularizer for the learned value functions, simultaneously ensuring training stability while enhancing the generalization capabilities of our policy gradient method—precisely the properties needed to bridge the simulation-to-reality gap in practical reinforcement learning applications.

3.2.2 Entropic Regularization and Sinkhorn Algorithm

While the Kantorovich duality provides theoretical elegance, the computational complexity of solving the optimal transport problem scales poorly with dimension. We use entropic regularization to transform the original problem into

$$W_{c, \epsilon}(\mu, \nu) = \inf_{\gamma \in \Pi(\mu, \nu)} \left[\int c d\gamma + \epsilon \cdot \text{KL}(\gamma \| \mu \otimes \nu) \right] \quad (12)$$

where KL denotes the Kullback-Leibler divergence and $\epsilon > 0$ is the regularization parameter. The entropic regularization convexifies the primal problem and enables efficient computation through the celebrated Sinkhorn algorithm.

The entropic regularization can be understood as introducing a “logistical flexibility” term: rather than insisting on the absolute minimum cost regardless of planning complexity, we allow for slightly suboptimal routes in exchange for a more manageable and robust distribution plan. The $\text{KL}(\gamma \| \mu \otimes \nu)$ penalizes plans that deviate too drastically from simply independently sampling source and destination pairs, effectively smoothing the transport map.

By the entropic regularization, the dual formulation of this regularized problem becomes into the following form

$$W_{c, \epsilon}(\mu, \nu) = \sup_{f, g} \left[\int f d\mu + \int g d\nu - \epsilon \int \left(e^{\frac{f(x) + g(y) - c(x, y)}{\epsilon}} - 1 \right) d\mu(x) d\nu(y) \right] \quad (13)$$

The optimal solutions satisfy the elegant scaling relations, which actually form the principle of the Sinkhorn algorithm

$$f(x) = -\epsilon \log \int e^{\frac{g(y) - c(x,y)}{\epsilon}} d\nu(y), \quad g(y) = -\epsilon \log \int e^{\frac{f(x) - c(x,y)}{\epsilon}} d\mu(x) \quad (14)$$

The mathematical structure of Kantorovich duality and entropic regularization provides both computational tractability and interpretability: the potentials f and g correspond to value functions that assess state and action qualities, at the same time, the entropic regularization ensures stable learning and prevents overfitting to the demonstration data. In our DOT-PG framework, we integrate the Sinkhorn algorithm as an initialization method for the Critic network, which approximates the OT dual variable and significantly enhances learning efficiency and stability.

3.3 Reinforcement Learning Foundations

3.3.1 Markov Decision Process

The theoretical foundation of our approach is built upon the Markov Decision Process (MDP) framework, which provides a mathematical formalism for sequential decision-making problems. An MDP is formally defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma, \rho_0)$, where $\mathcal{S}$ is the State space representing all possible configurations of the environment; $\mathcal{A}$ is the Action space containing all possible decisions the agent can make; $\mathcal{P}(s'|s, a)$ is the State transition dynamics specifying the probability of transitioning to state s' from state s after taking action a ; $r(s, a)$ is the Reward function, providing immediate feedback for taking action a in state s ; $\gamma \in [0, 1)$ is the Discount factor, balancing immediate versus future rewards; $\rho_0(s)$ is the Initial state distribution. The agent's behavior is governed by a policy $\pi(a|s)$, which defines a probability distribution over actions given states. The objective in standard RL is to find an optimal policy π^* that maximizes the expected discounted return, which is usually written in the following mathematical form

$$J(\pi) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] \quad (15)$$

where $\tau = (s_0, a_0, s_1, a_1, \dots)$ denotes a trajectory sampled by following policy π which is actually a dynamic behavior. In our proposed framework, we maintain the MDP structure but eliminate the dependence on manually designed reward functions by leveraging the optimal transport duality to derive intrinsic learning signals from expert demonstrations.

3.3.2 State-Action Value Functions

The state-action value function (Q-function) plays a central role in value-based reinforcement learning methods. For a given policy π , the Q-function $Q_\pi(s, a)$ represents the expected discounted return when starting from state s , taking action a , and thereafter following policy π , which is formally written by the following equation

$$Q_\pi(s, a) = \mathbb{E}_{\tau \sim \pi} \left[\sum_{k=0}^{\infty} \gamma^k r(s_{t+k}, a_{t+k}) \mid s_t = s, a_t = a \right]. \quad (16)$$

This formulation captures the long-term value of taking a specific action in a given state while committing to the policy π for all future decisions. According to classical reinforcement learning theory, the optimal Q-function $Q^*(s, a)$, which corresponds to the best achievable performance from any policy, satisfies the Bellman optimality equation, i.e.,

$$Q^*(s, a) = r(s, a) + \gamma \mathbb{E}_{s' \sim \mathcal{P}(\cdot|s, a)} \left[\max_{a'} Q^*(s', a') \right]. \quad (17)$$

This recursive relationship forms the foundation of many RL algorithms, expressing that the optimal value equals the immediate reward plus the discounted value of the best possible future state. The fixed-point nature of this equation enables iterative solution methods like Q-learning. Also, an advantage function $A^\pi(s, a)$ can be defined to provide a relative measure of action quality by comparing each action's value against the average performance of the policy

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s) \quad (18)$$

where $V^\pi(s) = \mathbb{E}_{a \sim \pi(\cdot|s)} [Q^\pi(s, a)]$ is the state-value function representing the expected return from state s when following policy π . The advantage function has several important properties: 1. zero expectation: $\mathbb{E}_{a \sim \pi(\cdot|s)} [A^\pi(s, a)] = 0$ for all states s ; 2. policy gradient: $\nabla_\theta J(\theta) = \mathbb{E}_{s \sim d^\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) A^\pi(s, a)]$; 3. intuitive interpretability: positive values indicate better-than-average actions, while negative values indicate worse-than-average actions.

In DOT-PG, these concepts connect elegantly with optimal transport theory, where the advantage function naturally emerges from the dual formulation and inherits the Lipschitz continuity properties essential for robust policy learning, providing a dense learning signal that guides policy improvement without requiring explicit reward specification.

4 Problem Formulation and DOT-PG Algorithm

4.1 Problem Formulation

4.1.1 Markov Decision Process and Distribution Matching

We consider the standard reinforcement learning framework formalized as a Markov Decision Process (MDP), defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, r, \gamma, \rho_0)$. However, unlike the traditional RL settings, we operate under the learning from demonstrations (LfD) paradigm where the reward function $r(s, a)$ is unknown or difficult to specify. Instead, we are provided with expert demonstrations $\mathcal{D}_E = (s_i, a_i)_{i=1}^N$ sampled from the expert’s state-action distribution $\rho_E(s, a)$.

The fundamental objective is to find a policy π_θ that minimizes the discrepancy between its induced state-action distribution ρ_{π_θ} and the expert distribution ρ_E . We formulate this as a Wasserstein distance minimization problem

$$\min_{\theta} W(\rho_{\pi_\theta}, \rho_E) = \min_{\theta} \inf_{\gamma \in \Pi(\rho_{\pi_\theta}, \rho_E)} \int c((s, a), (s_E, a_E)) d\gamma((s, a), (s_E, a_E)) \quad (19)$$

where $\Pi(\rho_{\pi_\theta}, \rho_E)$ denotes the set of all couplings between ρ_{π_θ} and ρ_E , and $c((s, a), (s_E, a_E))$ is a cost function measuring the distance between state-action pairs. Then, the discounted state-action distribution ρ_π is defined as follows

$$\rho_\pi(s, a) = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t P(s_t = s, a_t = a | \pi, \mathcal{P}) \quad (20)$$

4.1.2 Kantorovich Duality-Based Reformulation

Now, by applying the Kantorovich-Rubinstein duality, we can transform the primal optimal transport problem into a more tractable min-max optimization problem which is rigorously given by the following formal expression

$$\min_{\theta} \max_{f \in \text{Lip}1} [\mathbb{E}_{(s,a) \sim \rho_E} [f(s, a)] - \mathbb{E}_{(s,a) \sim \rho_{\pi_\theta}} [f(s, a)]] \quad (21)$$

This dual reformulation offers significant computational advantages that facilitate practical implementation. Most importantly, the dual variable $f(s, a)$ can be efficiently approximated using neural networks, enabling the capture of complex value functions in high-dimensional state-action spaces. Furthermore, this parameterization allows for gradient-based optimization, making the framework compatible with stochastic gradient methods that scale to large datasets. Critically, the inherent Lipschitz constraint on $f(s, a)$ serves as a powerful regularizer that stabilizes the training process by preventing overfitting and ensuring smooth optimization landscapes, thereby simultaneously combining the representational capacity of deep learning with the theoretical robustness of optimal transport.

4.1.3 Connecting Optimal Transport with Policy Gradients

Traditionally, the foundation for policy-based reinforcement learning methods is established by a certain of policy gradient. That is, for a differentiable policy π_θ parameterized by θ , the gradient of the expected return is given by

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}} [\nabla_{\theta} \log \pi_{\theta}(a|s) Q^{\pi_{\theta}}(s, a)]. \quad (22)$$

The basic idea herein begins with the expression for the expected return which is written as the following form

$$J(\pi_{\theta}) = \mathbb{E}_{\tau \sim \pi_{\theta}} \left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t) \right] = \int \rho_{\pi_{\theta}}(s, a) r(s, a) ds da. \quad (23)$$

Taking the gradient and applying the log-derivative trick immediately gives the policy gradient as below

$$\nabla_{\theta} J(\pi_{\theta}) = \int \nabla_{\theta} \rho_{\pi_{\theta}}(s, a) r(s, a) ds da = \int \rho_{\pi_{\theta}}(s, a) \nabla_{\theta} \log \rho_{\pi_{\theta}}(s, a) r(s, a) ds da \quad (24)$$

Using the identity $\nabla_{\theta} \log \rho_{\pi_{\theta}}(s, a) = \nabla_{\theta} \log \pi_{\theta}(a|s) + \nabla_{\theta} \log d^{\pi_{\theta}}(s)$, where $d^{\pi_{\theta}}(s)$ is the discounted state distribution, and noting that the term involving $\nabla_{\theta} \log d^{\pi_{\theta}}(s)$ vanishes in expectation, we obtain the desired expression. Furthermore, to reduce variance in the procedure of gradient estimation, the classical RL method is revised to subtract a state-dependent baseline $V^{\pi_{\theta}}(s)$ from the action-value function, defining the advantage function $A^{\pi_{\theta}}(s, a) = Q^{\pi_{\theta}}(s, a) - V^{\pi_{\theta}}(s)$. This strategy is then developed to preserve the unbiasedness of the gradient estimate while reducing variance

$$\nabla_{\theta} J(\pi_{\theta}) = \mathbb{E}_{(s,a) \sim \rho_{\pi_{\theta}}} [\nabla_{\theta} \log \pi_{\theta}(a|s) A^{\pi_{\theta}}(s, a)]. \quad (25)$$

Our key theoretical contribution establishes a fundamental connection between distribution matching and policy optimization by extending the standard policy gradient to the optimal transport setting. Specifically, for deterministic policies, we demonstrate that the gradient of the 1-Wasserstein distance admits a policy gradient decomposition

$$\nabla_{\theta} W(\rho_{\pi_{\theta}}, \rho_E) = -\mathbb{E}_{s \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \pi_{\theta}(s) \nabla_a f^*(s, a) \Big|_{a=\pi_{\theta}(s)} \right] \quad (26)$$

where f^* is the optimal Kantorovich dual variable. This result reveals that $\nabla_a f^*(s, a)$ plays a role analogous to the advantage function $A^{\pi_{\theta}}(s, a)$ in standard policy gradient methods, providing directional guidance for policy updates to minimize distribution discrepancy. The negative sign reflects that we are minimizing the Wasserstein distance, in contrast to maximizing expected return in conventional RL. This theoretical insight enables reward-free policy optimization by directly matching the agent’s state-action distribution to expert demonstrations, bypassing the need for manual reward engineering while maintaining the benefits of policy gradient methods for stable optimization.

Theorem 3 (OT-Based Policy Gradient for Deterministic Policies). *Let π_{θ} be a deterministic policy, $d^{\pi_{\theta}}$ the discounted state visitation distribution, and ρ_E the expert state-action distribution. Under appropriate regularity conditions, the gradient of the 1-Wasserstein distance with respect to policy parameters is given by the following equation*

$$\nabla_{\theta} W(\rho_{\pi_{\theta}}, \rho_E) = -\mathbb{E}_{s \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \pi_{\theta}(s) \nabla_a f^*(s, a) \Big|_{a=\pi_{\theta}(s)} \right]$$

where $f^*(s, a)$ is the optimal Kantorovich dual function, $\rho_{\pi_{\theta}}(s, a) = d^{\pi_{\theta}}(s) \cdot \delta(a - \pi_{\theta}(s))$ for deterministic policies.

Proof. We start from the following formulation of the discussed Kantorovich duality and entropic regularization

$$W(\rho_{\pi_{\theta}}, \rho_E) = \sup_{f \in \text{Lip}_1} \left[\mathbb{E}_{\rho_E} [f(s, a)] - \mathbb{E}_{\rho_{\pi_{\theta}}} [f(s, a)] \right].$$

Let f^* denote the optimal dual function achieving this supremum. Under the envelope theorem and assuming sufficient regularity conditions for differentiation under the supremum, the gradient with respect to θ satisfies that

$$\nabla_{\theta} W(\rho_{\pi_{\theta}}, \rho_E) = -\nabla_{\theta} \mathbb{E}_{\rho_{\pi_{\theta}}} [f^*(s, a)]$$

where we note that the optimal f^* depends on θ only through the constraint of the supremum, and this dependence vanishes at optimality by the envelope theorem. For a deterministic policy π_{θ} , the state-action distribution factorizes as $\rho_{\pi_{\theta}}(s, a) = d^{\pi_{\theta}}(s) \cdot \delta(a - \pi_{\theta}(s))$, where $d^{\pi_{\theta}}$ is the discounted state visitation distribution. Therefore, we have

$$\mathbb{E}_{\rho_{\pi_{\theta}}} [f^*(s, a)] = \mathbb{E}_{s \sim d^{\pi_{\theta}}} [f^*(s, \pi_{\theta}(s))].$$

Applying the chain rule, $\nabla_{\theta} \mathbb{E}_{s \sim d^{\pi_{\theta}}} [f^*(s, \pi_{\theta}(s))] = \mathbb{E}_{s \sim d^{\pi_{\theta}}} [\nabla_{\theta} f^*(s, \pi_{\theta}(s))]$, where we have exchanged expectation and gradient under appropriate regularity conditions. Now, applying the chain rule to $\nabla_{\theta} f^*(s, \pi_{\theta}(s))$, $\nabla_{\theta} f^*(s, \pi_{\theta}(s)) = \nabla_{\theta} \pi_{\theta}(s) \cdot \nabla_a f^*(s, a) \Big|_{a=\pi_{\theta}(s)}$. Note that the term involving $\nabla_{\theta} f^*(s, a)$ is zero by the envelope theorem, as f^* is optimal with respect to the dual objective. Combining these results yields the following equation

$$\nabla_{\theta} W(\rho_{\pi_{\theta}}, \rho_E) = -\mathbb{E}_{s \sim d^{\pi_{\theta}}} \left[\nabla_{\theta} \pi_{\theta}(s) \cdot \nabla_a f^*(s, a) \Big|_{a=\pi_{\theta}(s)} \right]. \quad \square$$

4.2 DOT-PG Solution Framework

The DOT-PG algorithm is built on three core principles: distribution matching as its foundation, alternating optimization as its methodology, and the integration of theory with practice. This design stems from understanding imitation learning’s essence - systematically guiding the agent’s state-action distribution to progressively converge toward the expert distribution. The algorithm employs a three-stage alternating optimization framework where each phase has clear theoretical foundations and serves distinct functions, ensuring both theoretical soundness and practical effectiveness.

4.2.1 Three-Component Architecture

The proposed DOT-PG framework employs a composite architecture which is composed of the following networks

1. Policy Network (Actor) $\pi_{\theta}(a|s)$:

- Architecture: Multi-layer perceptron with residual connections
- Output: Parameters of diagonal Gaussian distribution $\mathcal{N}(\mu_{\theta}(s), \sigma_{\theta}(s))$
- Normalization: Layer normalization and tanh activations for output bounds

2. Dual Network (Critic) $f_\phi(s, a)$:

- Input: Concatenated state-action vectors
- Architecture: Spectral normalization for Lipschitz enforcement
- Constraint: Gradient penalty with $\lambda > 0$
- Role: Approximates optimal transport dual variable

3. Q-Network $Q_\psi(s, a)$:

- Purpose: Long-horizon value estimation
- Training: Temporal difference learning with target networks
- Usage: Stabilizes policy updates through advantage estimation

4.2.2 Alternating Optimization Strategy

Stage 1 - Critic Update: Given fixed policy, maximize the Kantorovich dual objective

$$\max_{\phi} [\mathbb{E} \rho_E[f_\phi] - \mathbb{E} \rho_\pi[f_\phi] - \lambda \mathbb{E} \hat{x}[(|\nabla f_\phi(\hat{x})|_2 - 1)^2]] \quad (27)$$

In the first stage of distribution measurement, based on the Kantorovich duality theory, we achieve accurate estimation of the Wasserstein distance through a discriminator network with gradient penalty. This design not only provides a smooth optimization landscape but also delivers meaningful gradient signals even when the distribution supports do not overlap, fundamentally resolving the training instability issues prevalent in traditional adversarial imitation learning.

Stage 2 - Value Learning: Update Q-function using modified Bellman equation

$$Q_\psi(s, a) \leftarrow f_\phi(s, a) + \gamma \mathbb{E}_{s' \sim \mathcal{P}}[Q_{\psi_{\text{target}}}(s', \pi_\theta(s'))] \quad (28)$$

The second stage of long-horizon value estimation extends the distribution matching signal obtained in the previous step to the temporal decision-making level. By using the Kantorovich dual variable as an intrinsic reward signal, we construct a modified Bellman equation that enables the Q-function to learn the value of state-action pairs in long-term distribution matching. This design connects instantaneous distribution matching with long-term policy optimization, while ensuring the stability of value estimation through techniques such as target networks and clipped double Q-learning.

Stage 3 - Policy Improvement: Update policy to minimize Wasserstein distance

$$\min_{\theta} \mathbb{E} s \sim \rho_\pi[Q_\psi(s, \pi_\theta(s))] \quad (29)$$

The third stage of policy optimization translates the OT policy gradient theorem into practice. Policy updates based on the reparameterization trick not only provide low-variance gradient estimates but also maintain necessary exploration capability through entropy regularization. This stage transforms the achievements of the first two stages into substantive policy improvements, directly optimizing the policy to minimize the Wasserstein distance to the expert distribution.

These three stages form an organic whole through sophisticated coordination mechanisms. Distribution measurement provides the signal foundation for value learning, value estimation offers long-term guidance for policy optimization, and policy improvements in turn influence subsequent distribution measurements, creating a virtuous optimization cycle. In practical implementation, we ensure training stability through differentiated update frequencies—the discriminator updates most frequently to guarantee the accuracy of distribution measurement, value function updates follow, policy updates are relatively cautious, while target networks track the changes of the main network at the slowest pace.

4.3 DOT-PG Algorithm and Design Rationale

A structured training procedure is implemented by integrating optimal transport theory with deep reinforcement learning, that operates through five coordinated phases to ensure stable and efficient learning from expert demonstrations. The DOT-PG algorithm is built around several key components that work in concert to enable effective imitation learning:

1. **Dual Network (Critic):** The network f_ϕ serves as the optimal transport dual variable, providing a measure of distribution discrepancy between agent and expert behaviors. The gradient penalty term ensures the Lipschitz continuity required by the Kantorovich-Rubinstein duality, enabling stable estimation of the Wasserstein distance. This approach provides superior stability compared to weight clipping methods commonly used in Wasserstein-based algorithms.
2. **Policy and Q-Networks:** The policy network π_θ and Q-network Q_ψ work together to optimize long-term behavior. The Q-function learns to estimate cumulative future “rewards” as defined by the dual network output, while the policy

network adjusts to maximize these estimated returns. This modular design maintains theoretical consistency with maximum entropy reinforcement learning frameworks.

3. Alternating Optimization: The algorithm is to employ a carefully sequenced update procedure where the dual network is optimized to accurately measure distribution distance, followed by policy improvements based on these measurements. This alternating approach prevents training instability while ensuring consistent progress. This structured approach is to prevent training instability while ensuring consistent progress toward policy optimization.

4. Target Networks: The use of target networks for Q-value estimation ($Q_{\psi_{\text{target}}}$) reduces the risk of divergent training behavior that can occur when using rapidly changing value estimates in temporal difference learning. A kind of soft update mechanism (τ -weighted averaging) is needed to ensure smooth target value transitions.

Algorithm 1 Dual Optimal Transport Policy Gradient (DOT-PG)

Require: Expert dataset $\mathcal{D}_E$, training steps T , batch size B

Ensure: Trained policy π_θ

Initialize: Policy network π_θ , dual network f_ϕ , Q-network Q_ψ , target network $Q_{\psi_{\text{target}}}$, replay buffer $\mathcal{B}$

for $t = 1$ to T **do**

Phase 1: Data Collection

Execute current policy π_θ in environment

Store collected transitions (s, a, s') in replay buffer $\mathcal{B}$

Phase 2: Dual Network Update

Sample expert batch $\mathcal{B}_E \sim \mathcal{D}_E$ and policy batch $\mathcal{B}_\pi \sim \mathcal{B}$

Compute interpolated points: $\hat{s} = \epsilon s_E + (1 - \epsilon)s_\pi$, $\hat{a} = \epsilon a_E + (1 - \epsilon)a_\pi$

Calculate gradient penalty: $\mathcal{L}_{GP} = \mathbb{E}[(\|\nabla f_\phi(\hat{s}, \hat{a})\|_2 - 1)^2]$

Update dual network: $\phi \leftarrow \phi - \alpha_\phi \nabla_\phi (\mathbb{E}[f_\phi(s, a)] - \mathbb{E}[f_\phi(s_E, a_E)] + \lambda \mathcal{L}_{GP})$

Phase 3: Q-Network Update

Compute target values: $y = f_\phi(s, a) + \gamma \mathbb{E}[Q_{\psi_{\text{target}}}(s', \pi_\theta(s'))]$

Update Q-network: $\psi \leftarrow \psi - \alpha_\psi \nabla_\psi \mathbb{E}[(Q_\psi(s, a) - y)^2]$

Phase 4: Policy Update

Update policy parameters: $\theta \leftarrow \theta + \alpha_\theta \nabla_\theta \mathbb{E}[Q_\psi(s, \pi_\theta(s))]$

Phase 5: Target Network Update

Update target network: $\psi_{\text{target}} \leftarrow \tau \psi + (1 - \tau) \psi_{\text{target}}$

end for

To summarize, our proposed DOT-PG’s five-phase optimization delivers distinctive practical advantages through several key characteristics. First, the algorithm ensures remarkable stability via its gradient penalty mechanism, which maintains Lipschitz continuity without compromising gradient flow, outperforming traditional weight clipping methods. Second, strategic batch reuse across phases maximizes data efficiency from both expert demonstrations and agent interactions, while the modular architecture enables flexible integration of diverse network architectures within the core optimal transport framework. Last and crucially, the implementation maintains rigorous theoretical consistency, with each component directly corresponding to established optimal transport and reinforcement learning principles, ensuring mathematical faithfulness while enabling effective imitation learning in continuous control domains.

4.4 Theoretical Guarantees for DOT-PG Solution

Theorem 4 (Optimal Transport Dual Convergence). *Let f_ϕ be the dual network parameterized by ϕ , and ρ_π, ρ_E the agent and expert state-action distributions respectively. Under gradient penalty regularization with coefficient λ , the sequence $\{\phi_k\}$ converges to the optimal Kantorovich dual variable, which is precisely expressed by*

$$\lim_{k \rightarrow \infty} f_{\phi_k} = f^* = \arg \max_{f \in \text{Lip}_1} [\mathbb{E}_{\rho_E}[f(s, a)] - \mathbb{E}_{\rho_\pi}[f(s, a)]] \quad (30)$$

with the gradient penalty ensuring L_f -Lipschitz continuity and superior stability compared to weight clipping methods.

Proof. The regularized dual optimization problem is given by the following form under gradient penalty regularization.

$$\max_{f \in \mathcal{F}} [\mathbb{E}_{\rho_E}[f(s, a)] - \mathbb{E}_{\rho_\pi}[f(s, a)] - \lambda \mathbb{E}_{\hat{\rho}}[(\|\nabla f\|_2 - 1)^2]] \quad (31)$$

where $\mathcal{F}$ denotes the neural network function class with sufficient expressive power (deep networks can approximate continuous functions), $\hat{\rho}$ represents the interpolation distribution, defined as uniform sampling along straight lines between ρ_π and ρ_E . Specifically, for pairs $(s, a) \sim \rho_\pi$ and $(s_E, a_E) \sim \rho_E$, interpolation points are $x_\epsilon = \epsilon(s, a) +$

$(1 - \epsilon)(s_E, a_E)$ with $\epsilon \sim U[0, 1]$. It can be clearly seen from previous discussion that this formulation combines the Kantorovich dual objective with a gradient penalty term that enforces approximate 1-Lipschitz continuity.

Next, let us analyze the gradient penalty properties. The term $\lambda \mathbb{E}_{\hat{\rho}}[(\|\nabla f\|_2 - 1)^2]$ ensures Lipschitz continuity by

$$\mathbb{E}_{\hat{\rho}}[(\|\nabla f\|_2 - 1)^2] = 0 \Rightarrow \|\nabla f\|_2 = 1 \text{ almost everywhere} \Rightarrow |f(x) - f(y)| \leq \|x - y\|_2, \forall x, y. \quad (32)$$

In practice, minimizing this penalty encourages $\|\nabla f\|_2 \approx 1$, achieving approximate 1-Lipschitz continuity. Because the neural networks in $\mathcal{F}$ are smooth functions, and the gradient penalty covers critical regions between distributions, thus the interpolation distribution $\hat{\rho}$ ensures comprehensive coverage. Compared to weight clipping, gradient penalty preserves network capacity while providing more stable optimization by avoiding gradient vanishing/explosion issues.

Now, consider the convergence analysis for the stochastic gradient descent update for dual network parameters, i.e.,

$$\phi_{k+1} = \phi_k + \alpha_{\phi,k} \nabla_{\phi} (\mathbb{E}_{\rho_E}[f_{\phi}] - \mathbb{E}_{\rho_{\pi}}[f_{\phi}] - \lambda \mathbb{E}_{\hat{\rho}}[(\|\nabla f_{\phi}\|_2 - 1)^2]).$$

The learning rates $\alpha_{\phi,k}$ satisfy Robbins-Monro conditions $\sum_{k=1}^{\infty} \alpha_{\phi,k} = \infty$, $\sum_{k=1}^{\infty} \alpha_{\phi,k}^2 < \infty$. These conditions ensure that learning rates are sufficiently small to suppress noise but large enough to guarantee convergence.

Note that the optimization objective is concave because the expectation terms are linear in f and the negative gradient penalty term is concave (as it's the negative of a convex function). Given sufficient capacity in $\mathcal{F}$ (neural networks can represent optimal f^*), stochastic approximation convergence theorems (Kushner & Yin, 2003) guarantee that $\{\phi_k\}$ converges almost surely to the stationary point corresponding to f^* . Formally, define $L(\phi) = \mathbb{E}_{\rho_E}[f_{\phi}] - \mathbb{E}_{\rho_{\pi}}[f_{\phi}] - \lambda \mathbb{E}_{\hat{\rho}}[(\|\nabla f_{\phi}\|_2 - 1)^2]$. Since $L(\phi)$ is concave and Lipschitz continuous, the stochastic gradient descent updates converge to $\arg \max L(\phi)$, which equals the Kantorovich dual solution as the gradient penalty term approaches zero at optimum. Under gradient penalty regularization, the dual network sequence $\{\phi_k\}$ converges almost surely to the optimal Kantorovich dual variable f^* , with gradient penalty ensuring stable and efficient optimization. $\square$

Theorem 5 (Wasserstein Distance Estimation Stability). *The stable estimation of the 1-Wasserstein distance is provided*

$$W(\rho_{\pi}, \rho_E) = \mathbb{E}_{\rho_E}[f^*] - \mathbb{E}_{\rho_{\pi}}[f^*] + O(\lambda^{-1}) \quad (33)$$

where the estimation error decays exponentially with the gradient penalty strength λ in regularized dual optimization.

Proof. The Wasserstein distance is given by the Kantorovich-Rubinstein duality theorem with 1-Wasserstein distance

$$W(\rho_{\pi}, \rho_E) = \sup_{f \in \text{Lip}_1} [\mathbb{E}_{\rho_E}[f(s, a)] - \mathbb{E}_{\rho_{\pi}}[f(s, a)]] \quad (34)$$

can be optimized over 1-Lipschitz functions, where the supremum is achieved by the optimal Kantorovich potential f^* .

According to the Kantorovich regularized estimation, we solve a regularized version of this optimization problem, i.e.,

$$\hat{W} = \mathbb{E}_{\rho_E}[f_{\lambda}^*] - \mathbb{E}_{\rho_{\pi}}[f_{\lambda}^*] \quad (35)$$

where f_{λ}^* is the solution to the gradient-penalized optimization problem which is mathematically expressed by

$$f_{\lambda}^* = \arg \max_{f \in \mathcal{F}} [\mathbb{E}_{\rho_E}[f(s, a)] - \mathbb{E}_{\rho_{\pi}}[f(s, a)] - \lambda \mathbb{E}_{\hat{\rho}}[(\|\nabla f\|_2 - 1)^2]].$$

Next, we will show that this regularization ensures practical computability while maintaining theoretical guarantees. In fact, the estimation error $|W - \hat{W}|$ can be bounded by applying the following relaxation techniques of norm computation

$$|W - \hat{W}| = |(\mathbb{E}_{\rho_E}[f^*] - \mathbb{E}_{\rho_{\pi}}[f^*]) - (\mathbb{E}_{\rho_E}[f_{\lambda}^*] - \mathbb{E}_{\rho_{\pi}}[f_{\lambda}^*])| = |\mathbb{E}_{\rho_E}[f^* - f_{\lambda}^*] - \mathbb{E}_{\rho_{\pi}}[f^* - f_{\lambda}^*]| \quad (36)$$

$$\leq |\mathbb{E}_{\rho_E}[f^* - f_{\lambda}^*]| + |\mathbb{E}_{\rho_{\pi}}[f^* - f_{\lambda}^*]| \leq \|f^* - f_{\lambda}^*\|_{\infty} + \|f^* - f_{\lambda}^*\|_{\infty} = 2\|f^* - f_{\lambda}^*\|_{\infty} \quad (37)$$

because the expectations are taken over probability measures, the worst-case difference bounds the expected difference.

For the regularization error bound, using regularization theory for constrained optimization problems, it implies that

$$\|f^* - f_{\lambda}^*\|_{\infty} \leq C\lambda^{-1} \quad (38)$$

where the constant $C > 0$ depends on the smoothness properties of the distributions ρ_{π} and ρ_E , the geometric structure of the state-action space and the capacity and architecture of the neural network function class $\mathcal{F}$. This bound of the approximation error follows from the fact that the gradient penalty term acts as a Tikhonov regularizer, and the regularization error typically decays polynomially with the regularization parameter for well-posed problems.

Motivated by Kushner & Yin, 2003, the exponential decay $\|f^* - f_{\lambda}^*\|_{\infty} \leq C \exp(-c\lambda)$ is achieved when: the neural network has sufficient capacity to represent the optimal Kantorovich potential; or the distributions ρ_{π} and ρ_E have bounded support and satisfy certain smoothness conditions; the optimization algorithm effectively minimizes the gradient penalty term; the interpolation distribution $\hat{\rho}$ adequately covers the region between ρ_{π} and ρ_E . Combining these results, we obtain the final error bound $W(\rho_{\pi}, \rho_E) = \mathbb{E}_{\rho_E}[f_{\lambda}^*] - \mathbb{E}_{\rho_{\pi}}[f_{\lambda}^*] + O(\lambda^{-1})$ with the potential for exponential convergence $O(\exp(-c\lambda))$ under sufficient representational capacity. $\square$

Theorem 6 (Maximum Entropy Policy Optimization). *The policy network π_θ and Q -network Q_ψ maintain theoretical consistency with maximum entropy RL. To be precise, the optimal policy satisfies the following equation*

$$\pi^*(a|s) = \frac{\exp(Q^*(s, a) - V^*(s))}{\int \exp(Q^*(s, a') - V^*(s)) da'} \quad (39)$$

where $Q^*(s, a)$ represents the cumulative expected output of the dual network f^* , i.e., the immediate reward.

Proof. Considering the maximum entropy objective formulation incorporating an entropy bonus in the standard RL

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t (f(s_t, a_t) + \alpha H(\pi(\cdot|s_t))) \right],$$

where $f(s_t, a_t)$ is the immediate objective/reward, $H(\pi(\cdot|s)) = -\int \pi(a|s) \log \pi(a|s) da$ is the policy entropy, $\alpha > 0$ is the temperature parameter that control the entropy regularization and $\gamma \in [0, 1]$ is the discount factor.

We consider the following soft Bellman equation of maximum entropy RL, defining the soft Q -function recursively, i.e.,

$$Q(s, a) = f(s, a) + \gamma \mathbb{E}_{s' \sim P(\cdot|s, a)} [V(s')],$$

where the soft value function $V(s)$ represents the expected maximum entropy return from state s , which is given by

$$V(s) = \alpha \log \int \exp \left(\frac{Q(s, a)}{\alpha} \right) da$$

where α is the temperature parameter is introduced to control the exploration-exploitation tradeoff: if $\alpha \rightarrow 0$, then the solution approaches a deterministic policy, and if $\alpha \rightarrow \infty$, the solution approaches a total uniform random policy.

To analyze the optimal policy characterization of the policy, note that it maximizes the expected return plus entropy

$$\pi^* = \arg \max_{\pi} \mathbb{E}_\pi [Q(s, a) - \alpha \log \pi(a|s)] = \arg \max_{\pi} \int \pi(a|s) (Q(s, a) - \alpha \log \pi(a|s)) da,$$

treating the policy optimization as a regularized expected reward maximization with the entropy term as a regularizer.

Essentially, we solve this variational optimization problem under the normalization constraint $\int \pi(a|s) da = 1$, that is,

$$\mathcal{L} = \int \pi(a|s) Q(s, a) da - \alpha \int \pi(a|s) \log \pi(a|s) da + \lambda \left(1 - \int \pi(a|s) da \right).$$

Then the Euler-Lagrange equation is used from calculus of variations and take the functional derivative w.r.t. $\pi(a|s)$

$$\frac{\partial \mathcal{L}}{\partial \pi(a|s)} = Q(s, a) - \alpha (\log \pi(a|s) + 1) - \lambda = 0.$$

To find the optimal policy solution, it suffices to solve the above equation $Q(s, a) - \alpha (\log \pi(a|s) + 1) - \lambda = 0$ with

$$\pi^*(a|s) = \exp \left(\frac{Q(s, a) - \lambda - \alpha}{\alpha} \right).$$

To satisfy the normalization constraint, we compute the partition function by solving for the normalization constant

$$\int \pi^*(a|s) da = \int \exp \left(\frac{Q(s, a) - \lambda - \alpha}{\alpha} \right) da = 1, \quad \exp \left(\frac{\lambda + \alpha}{\alpha} \right) = \int \exp \left(\frac{Q(s, a)}{\alpha} \right) da.$$

Finally, recognizing that the soft value function $V(s) = \alpha \log \int \exp \left(\frac{Q(s, a)}{\alpha} \right) da$ gives directly the following equation

$$\pi^*(a|s) = \frac{\exp(Q(s, a)/\alpha)}{\int \exp(Q(s, a')/\alpha) da'} = \frac{\exp(Q(s, a) - V(s))}{\int \exp(Q(s, a') - V(s)) da'}. \quad \square$$

Theorem 7 (Long-term Behavior Optimization). *Under the modular policy- Q network architecture, the learned policy optimizes long-term distribution matching, which is precisely described by the following equation*

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{s \sim \rho_\pi} [\nabla_\theta D_{KL}(\pi_\theta(\cdot|s) \parallel \pi^*(\cdot|s))] + O(\epsilon_Q) \quad (40)$$

where the term ϵ_Q represents the Q -function approximation error defined by $\epsilon_Q = \max_{s, a} |Q_\psi(s, a) - Q^*(s, a)|$.

Proof. Consider the policy gradient in maximum entropy RL with ρ_π as the state visitation distribution under policy π_θ

$$J(\pi_\theta) = \mathbb{E}_{s \sim \rho_\pi, a \sim \pi_\theta} [Q_\psi(s, a) - \alpha \log \pi_\theta(a|s)]$$

where the policy gradient is given by $\nabla_\theta J(\pi_\theta) = \mathbb{E}_{s \sim \rho_\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) (Q_\psi(s, a) - \alpha \log \pi_\theta(a|s))]$.

For the optimal policy reference, from the Maximum Entropy Policy Optimization theorem, it directly implies that

$$\pi^*(a|s) = \frac{\exp(Q^*(s, a)/\alpha)}{\int \exp(Q^*(s, a')/\alpha) da'}, \quad Q^*(s, a) = \alpha \log \pi^*(a|s) + V^*(s)$$

in which $Q^*(s, a)$ is the optimal Q-function and $V^*(s) = \alpha \log \int \exp(Q^*(s, a')/\alpha) da'$ is the optimal value function.

Considering that $Q_\psi(s, a) = Q^*(s, a) + \epsilon_Q(s, a)$ where $\epsilon_Q(s, a)$ represents the Q-function approximation error, then

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{s \sim \rho_\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) (Q^*(s, a) + \epsilon_Q(s, a) - \alpha \log \pi_\theta(a|s))]$$

Recall the definition of KL divergence is given by the equation $D_{KL}(\pi_\theta \| \pi^*) = \mathbb{E}_{a \sim \pi_\theta} [\log \pi_\theta(a|s) - \log \pi^*(a|s)]$. Taking the gradient with respect to θ and note that $\mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s)] = \nabla_\theta \int \pi_\theta(a|s) da = \nabla_\theta 1 = 0$, we have

$$\begin{aligned} \nabla_\theta D_{KL}(\pi_\theta \| \pi^*) &= \mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) (\log \pi_\theta(a|s) - \log \pi^*(a|s)) + \nabla_\theta \log \pi_\theta(a|s)] \\ &= \mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) (\log \pi_\theta(a|s) - \log \pi^*(a|s))]. \end{aligned}$$

By connecting policy gradient and KL divergence and substituting $Q^*(s, a) = \alpha \log \pi^*(a|s) + V^*(s)$, it implies that

$$\begin{aligned} \nabla_\theta J(\pi_\theta) &= \alpha \mathbb{E}_{s \sim \rho_\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) (\log \pi^*(a|s) - \log \pi_\theta(a|s))] \\ &\quad + \mathbb{E}_{s \sim \rho_\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) (V^*(s) + \epsilon_Q(s, a))]. \end{aligned}$$

Note that $\alpha \mathbb{E}_{s \sim \rho_\pi} [\nabla_\theta D_{KL}(\pi_\theta(\cdot|s) \| \pi^*(\cdot|s))]$, $\mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) V^*(s)] = V^*(s) \mathbb{E}_{a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s)] = 0$. Therefore, we're left with $\nabla_\theta J(\pi_\theta) = \alpha \mathbb{E}_{s \sim \rho_\pi} [\nabla_\theta D_{KL}(\pi_\theta(\cdot|s) \| \pi^*(\cdot|s))] + \mathbb{E}_{s \sim \rho_\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) \epsilon_Q(s, a)]$.

In the above equation, the error term $O(\epsilon_Q) = \mathbb{E}_{s \sim \rho_\pi, a \sim \pi_\theta} [\nabla_\theta \log \pi_\theta(a|s) \epsilon_Q(s, a)]$ represents the impact of Q-function approximation error on the policy gradient. If ϵ_Q is small (accurate Q-function approximation), this term becomes negligible. Assuming the policy gradient and Q-error are bounded, we can express this as $O(\epsilon_Q)$, where ϵ_Q represents the maximum Q-function approximation error, that is, $\epsilon_Q = \max_{s, a} |Q_\psi(s, a) - Q^*(s, a)|$. Absorbing the constant α into the definition (or assuming $\alpha = 1$ for simplicity), it implies that the final conclusion is achieved

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_{s \sim \rho_\pi} [\nabla_\theta D_{KL}(\pi_\theta(\cdot|s) \| \pi^*(\cdot|s))] + O(\epsilon_Q). \quad \square$$

Theorem 8 (Alternating Optimization Stability). *The carefully sequenced update procedure—dual network optimization followed by policy improvement—ensures training stability, which is rigorously described by the following inequality*

$$\mathbb{E}[W(\rho_{\pi_{k+1}}, \rho_E) - W(\rho_{\pi_k}, \rho_E)] \leq -\alpha \|\nabla W\|^2 + \beta \delta_k \quad (41)$$

where δ_k is the dual network approximation error at iteration k , and $\alpha, \beta > 0$ are two positive constants.

Proof. The alternating optimization sequence is composed of two updates driven by the gradients of $L(\phi, \theta)$, $f_{\phi_{k+1}}$

$$\phi_{k+1} = \phi_k + \alpha_\phi \nabla_\phi L(\phi_k, \theta_k), \quad \theta_{k+1} = \theta_k - \alpha_\theta \nabla_\theta W(\rho_{\pi_{\theta_k}}, \rho_E; f_{\phi_{k+1}}),$$

where $L(\phi, \theta)$ serves the dual objective for estimating $W(\rho_\pi, \rho_E)$, the dual network $f_{\phi_{k+1}}$ serves the policy optimization.

By performance difference decomposition, we analyze the expected change in true Wasserstein distance, that is,

$$\begin{aligned} \mathbb{E}[W(\rho_{\pi_{k+1}}, \rho_E) - W(\rho_{\pi_k}, \rho_E)] &= \underbrace{\mathbb{E}[W(\rho_{\pi_{k+1}}, \rho_E) - W(\rho_{\pi_{k+1}}, \rho_E; f_{\phi_{k+1}})]}_{\text{Term A: Dual approximation error for } \pi_{k+1}} \\ &\quad + \underbrace{\mathbb{E}[W(\rho_{\pi_{k+1}}, \rho_E; f_{\phi_{k+1}}) - W(\rho_{\pi_k}, \rho_E; f_{\phi_{k+1}})]}_{\text{Term B: Policy improvement}} + \underbrace{\mathbb{E}[W(\rho_{\pi_k}, \rho_E; f_{\phi_{k+1}}) - W(\rho_{\pi_k}, \rho_E)]}_{\text{Term C: Dual improvement for } \pi_k}. \end{aligned}$$

For Term A: by the dual formulation of Wasserstein distance and the approximation properties, it can be seen that

$$\mathbb{E}[W(\rho_{\pi_{k+1}}, \rho_E) - W(\rho_{\pi_{k+1}}, \rho_E; f_{\phi_{k+1}})] \leq \mathbb{E}[\|f^* - f_{\phi_{k+1}}\|_\infty] = \delta_k, \quad \delta_k = \|f_{\phi_{k+1}} - f_{\theta_k}^*\|_\infty.$$

This term represents how well our dual network approximates the true optimal transport cost for the new policy.

For Term B: using the policy gradient theorem and Taylor expansion, this term is analyzed by the following equation

$$\begin{aligned} W(\rho_{\pi_{k+1}}, \rho_E; f_{\phi_{k+1}}) - W(\rho_{\pi_k}, \rho_E; f_{\phi_{k+1}}) &= \langle \nabla_\theta W(\rho_{\pi_k}, \rho_E; f_{\phi_{k+1}}), \theta_{k+1} - \theta_k \rangle + O(\|\theta_{k+1} - \theta_k\|^2) \\ &= -\alpha_\theta \|\nabla_\theta W(\rho_{\pi_k}, \rho_E; f_{\phi_{k+1}})\|^2 + O(\alpha_\theta^2) \leq -\alpha \|\nabla W_1\|^2 + O(\alpha_\theta^2), \quad \alpha = \alpha_\theta, \|\nabla W\| = \|\nabla_\theta W(\rho_{\pi_k}, \rho_E; f_{\phi_{k+1}})\|. \end{aligned}$$

For Term C: this term captures how the dual network update affects the Wasserstein estimate for the previous policy

$$\mathbb{E}[W(\rho_{\pi_k}, \rho_E; f_{\phi_{k+1}}) - W(\rho_{\pi_k}, \rho_E)] \leq -\gamma \mathbb{E}[\|f_{\phi_{k+1}} - f_{\phi_k}\|^2] \leq 0.$$

Then, by summing up all terms together, the following inequality holds which estimates $\mathbb{E}[W(\rho_{\pi_{k+1}}, \rho_E) - W(\rho_{\pi_k}, \rho_E)]$

$$\mathbb{E}[W(\rho_{\pi_{k+1}}, \rho_E) - W(\rho_{\pi_k}, \rho_E)] \leq \delta_k + (-\alpha \|\nabla W\|^2 + O(\alpha_\theta^2)) \leq -\alpha \|\nabla W\|^2 + \beta \delta_k + O(\alpha_\theta^2).$$

For error propagation analysis, by using the contraction property of the dual update, we analyze the term δ_{k+1} as below

$$\begin{aligned} \delta_{k+1} &= \|f_{\phi_{k+2}} - f_{\theta_{k+1}}^*\|_\infty \leq \|f_{\phi_{k+2}} - f_{\theta_k}^*\|_\infty + \|f_{\theta_k}^* - f_{\theta_{k+1}}^*\|_\infty \\ &\leq (1 - c\alpha_\phi)\delta_k + L\|\theta_{k+1} - \theta_k\| + O(\alpha_\phi^2) \leq (1 - c\alpha_\phi)\delta_k + O(\alpha_\theta) + O(\alpha_\phi^2) \end{aligned}$$

where $c > 0$ is the contraction rate, L is the Lipschitz constant of the optimal dual function with policy parameters. What's more, under appropriate learning rates satisfying: α_θ is small enough to make $O(\alpha_\theta^2)$ negligible, α_ϕ is chosen such that $(1 - c\alpha_\phi) < 1$ and the sequence $\{\delta_k\}$ converges to 0, we obtain the following monotonic improvement

$$\lim_{k \rightarrow \infty} \mathbb{E}[W_1(\rho_{\pi_{k+1}}, \rho_E) - W_1(\rho_{\pi_k}, \rho_E)] \leq 0. \quad \square$$

Theorem 9 (Two-Time-Scale Convergence). *Under Robbins-Monro conditions for learning rates $\alpha_\phi \gg \alpha_\theta$, the alternating optimization converges, $\lim_{k \rightarrow \infty} \|\nabla_\theta W(\rho_{\pi_{\theta_k}}, \rho_E)\| = 0$ almost surely, where the critic learning system is designed to be faster than the actor system to ensure stable policy updates.*

Proof. We consider the coupled stochastic approximation system which is composed of updated parameters θ_k and ϕ_k

$$\theta_{k+1} = \theta_k - \alpha_{\theta,k} \hat{g}_k, \quad \phi_{k+1} = \phi_k + \alpha_{\phi,k} \hat{h}_k$$

where $\hat{g}_k = \nabla_\theta W(\rho_{\pi_{\theta_k}}, \rho_E; \phi_k) + M_{\theta,k}$, $\hat{h}_k = \nabla_\phi L(\phi_k, \theta_k) + M_{\phi,k}$, $M_{\theta,k}$, $M_{\phi,k}$ are martingale difference sequences.

For learning rate conditions and time-scale separation, consider the Robbins-Monro conditions on learning rate, i.e.,

$$\sum_{k=0}^{\infty} \alpha_{\theta,k} = \infty, \quad \sum_{k=0}^{\infty} \alpha_{\theta,k}^2 < \infty, \quad \sum_{k=0}^{\infty} \alpha_{\phi,k} = \infty, \quad \sum_{k=0}^{\infty} \alpha_{\phi,k}^2 < \infty, \quad \lim_{k \rightarrow \infty} \frac{\alpha_{\theta,k}}{\alpha_{\phi,k}} = 0$$

is imposed. Here, this time-scale separation means that the critic (ϕ) evolves on a faster time-scale than the actor (θ).

Fast Subsystem Analysis (Critic Convergence) For fixed θ , consider the critic ODE, i.e., $\frac{d\phi}{dt} = \nabla_\phi L(\phi, \theta)$. Assume that for each fixed θ , $L(\phi, \theta)$ is strongly concave in ϕ with unique maximizer $\phi^*(\theta)$ and the stochastic gradient $\hat{h}_k$ satisfies: $\mathbb{E}[\hat{h}_k | \mathcal{F}_k] = \nabla_\phi L(\phi_k, \theta_k)$, $\mathbb{E}[\|\hat{h}_k\|^2 | \mathcal{F}_k] \leq C(1 + \|\phi_k\|^2 + \|\theta_k\|^2)$. By Theorem 2.2 of Borkar (2008) on two-time-scale stochastic approximation, for fixed θ , the fast subsystem converges: $\lim_{k \rightarrow \infty} \|\phi_k - \phi^*(\theta)\| = 0$ a.s..

Slow Subsystem Analysis (Actor Dynamics) On the slow time-scale, the actor update can be written as follows

$$\theta_{k+1} = \theta_k - \alpha_{\theta,k} [\nabla_\theta W_1(\rho_{\pi_{\theta_k}}, \rho_E; \phi^*(\theta_k)) + \epsilon_k + M_{\theta,k}]$$

where $\epsilon_k = \nabla_\theta W_1(\rho_{\pi_{\theta_k}}, \rho_E; \phi_k) - \nabla_\theta W(\rho_{\pi_{\theta_k}}, \rho_E; \phi^*(\theta_k))$ is the critic approximation error. From the fast subsystem convergence and the Lipschitz continuity of $\nabla_\theta W$, then it can be immediately derived that $\lim_{k \rightarrow \infty} \|\epsilon_k\| = 0$ a.s..

Consider an associated ODE of the stability, the limiting ODE for the slow subsystem is $\frac{d\theta}{dt} = -\nabla_\theta W(\rho_{\pi_\theta}, \rho_E; \phi^*(\theta))$. Assume that the function $V(\theta) = W(\rho_{\pi_\theta}, \rho_E)$ is continuously differentiable and serves as a Lyapunov function for the ODE. Compute the time derivative: $\frac{dV(\theta(t))}{dt} = -\langle \nabla_\theta W(\rho_{\pi_\theta}, \rho_E), \nabla_\theta W(\rho_{\pi_\theta}, \rho_E; \phi^*(\theta)) \rangle$. For the critic is sufficiently accurate such that: $\langle \nabla_\theta W_1(\rho_{\pi_\theta}, \rho_E), \nabla_\theta W_1(\rho_{\pi_\theta}, \rho_E; \phi^*(\theta)) \rangle \geq c \|\nabla_\theta W_1(\rho_{\pi_\theta}, \rho_E)\|^2$ for some $c > 0$. This ensures:

$$\frac{dV(\theta(t))}{dt} \leq -c \|\nabla_\theta W(\rho_{\pi_\theta}, \rho_E)\|^2 \leq 0.$$

Next, let us perform the martingale noise analysis, note that the martingale difference sequence $M_{\theta,k}$ satisfies: $\mathbb{E}[M_{\theta,k} | \mathcal{F}_k] = 0$, $\mathbb{E}[\|M_{\theta,k}\|^2 | \mathcal{F}_k] \leq K(1 + \|\theta_k\|^2)$. By the martingale convergence theorem and the square-summability of $\alpha_{\theta,k}$, it can be seen from the above conclusions that the cumulative noise effect vanishes.

Finally, we apply the Kushner-Clark lemma (Theorem 2.3.1 in Kushner & Yin, 2003) for two-time-scale systems, at the same time, under the stated assumptions and learning rate conditions, the iterates (θ_k, ϕ_k) converge almost surely to the set: $\{(\theta, \phi) : \phi = \phi^*(\theta), \nabla_\theta W(\rho_{\pi_\theta}, \rho_E; \phi^*(\theta)) = 0\}$. Moreover, by the accuracy of the critic approximation, $\nabla_\theta W(\rho_{\pi_\theta}, \rho_E; \phi^*(\theta)) = 0$, which means that $\nabla_\theta W(\rho_{\pi_\theta}, \rho_E) = 0$. Therefore, we conclude that

$$\lim_{k \rightarrow \infty} \|\nabla_\theta W(\rho_{\pi_{\theta_k}}, \rho_E)\| = 0 \quad \text{almost surely.} \quad \square$$

Theorem 10 (Target Network Stabilization). *The use of target networks $Q_{\psi_{\text{target}}}$ with soft update parameter τ ensures bounded temporal difference error, the norm bound of which is generally estimated by the following inequality*

$$\mathbb{E}[\|Q_\psi - Q^*\|] \leq \frac{C}{1 - \gamma(1 - \tau)} + O(\tau^{-1}) \quad (42)$$

where the constant C depends on the Lipschitz constants of the value functions and policy.

Proof. Let us define Q^* as Optimal Q-function, Q^k as current Q-function estimate at iteration k , Q_{target}^k as target Q-function at iteration k , τ as soft update parameter ($0 < \tau \ll 1$) and γ as discount factor as ($0 \leq \gamma < 1$). Then,

$$Q^{k+1} = Q^k + \alpha \delta_t \nabla Q^k(s_t, a_t), \quad Q_{\text{target}}^{k+1} = \tau Q^k + (1 - \tau) Q_{\text{target}}^k$$

are the update equations are receptively used for Q-function and target network. The TD error with target network is:

$$\delta_t = f(s_t, a_t) + \gamma Q_{\text{target}}^k(s_{t+1}, a_{t+1}) - Q^k(s_t, a_t) = (T^* Q_{\text{target}}^k - Q^k) + \epsilon_{\text{approx}}$$

where T^* is the Bellman optimality operator and ϵ_{approx} represents the related function approximation error. Define the estimation errors: $e^k = Q^k - Q^*$ (current network error) and $e_{\text{target}}^k = Q_{\text{target}}^k - Q^*$ (target network error). The Q-update is expressed as: $Q^{k+1} = Q^k + \alpha(T^* Q_{\text{target}}^k - Q^k + \epsilon_k)$ where ϵ_k encompasses both approximation and sampling errors.

$$e^{k+1} = (1 - \alpha)e^k + \alpha(T^* Q_{\text{target}}^k - Q^*) + \alpha\epsilon_k = (1 - \alpha)e^k + \alpha(T^*(Q^* + e_{\text{target}}^k) - Q^*) + \alpha\epsilon_k.$$

From the contraction property of T^* : $\|T^* Q - T^* Q'\| \leq \gamma \|Q - Q'\|$, $\|e^{k+1}\| \leq (1 - \alpha)\|e^k\| + \alpha\gamma\|e_{\text{target}}^k\| + \alpha\|\epsilon_k\|$. The target network update gives $e_{\text{target}}^{k+1} = \tau Q^k + (1 - \tau) Q_{\text{target}}^k - Q^* = \tau e^k + (1 - \tau) e_{\text{target}}^k$, which is considered by

$$\begin{bmatrix} \|e^{k+1}\| \\ \|e_{\text{target}}^{k+1}\| \end{bmatrix} \leq \begin{bmatrix} 1 - \alpha & \alpha\gamma \\ \tau & 1 - \tau \end{bmatrix} \begin{bmatrix} \|e^k\| \\ \|e_{\text{target}}^k\| \end{bmatrix} + \begin{bmatrix} \alpha\|\epsilon_k\| \\ 0 \end{bmatrix}.$$

Let $A = \begin{bmatrix} 1 - \alpha & \alpha\gamma \\ \tau & 1 - \tau \end{bmatrix}$. The spectral radius $\rho(A)$ determines stability, whose characteristic equation is given by

$$\lambda^2 - (2 - \alpha - \tau)\lambda + (1 - \alpha)(1 - \tau) - \alpha\gamma\tau = 0.$$

The eigenvalues satisfy $|\lambda| < 1$ when $(1 - \alpha)(1 - \tau) - \alpha\gamma\tau < 1$ which holds for $0 < \alpha, \tau < 1$ and $\gamma < 1$.

Assuming the approximation error is bounded ($\|\epsilon_k\| \leq \epsilon$), the steady-state solution satisfies the following inequality

$$(I - A) \begin{bmatrix} e_\infty \\ e_{\text{target}, \infty} \end{bmatrix} \leq \begin{bmatrix} \alpha\epsilon \\ 0 \end{bmatrix}, \quad e_\infty \leq \frac{\alpha\epsilon}{\alpha(1 - \gamma) - \alpha\gamma\tau + \alpha\tau} = \frac{\epsilon}{1 - \gamma + \gamma\tau + \tau}.$$

For small τ , we can approximate: $e_\infty \leq \frac{\epsilon}{1 - \gamma(1 - \tau)} + O(\tau^{-1})$. The constant C depends on the Lipschitz constant L_Q of the Q-function, the Lipschitz constant L_π of the policy and the Maximum approximation error ϵ , i.e., $C = L_Q L_\pi \epsilon$.

$$\mathbb{E}[\|Q_\psi - Q^*\|] \leq \frac{L_Q L_\pi \epsilon}{1 - \gamma(1 - \tau)} + O(\tau^{-1}) + O\left(\frac{\alpha}{\tau}\right).$$

The $O(\tau^{-1})$ term arises when τ is very small, while the $O(\alpha/\tau)$ term is the coupling between learning rates. $\square$

Theorem 11 (Smooth Value Transition). *The τ -weighted averaging mechanism provides smooth target value transitions:*

$$\|Q_{\psi_{\text{target}}}^{k+1} - Q_{\psi_{\text{target}}}^k\| \leq \tau \|Q_\psi^k - Q_{\psi_{\text{target}}}^k\| \quad (43)$$

which is able to prevent the divergent training behavior from rapidly changing value estimates.

Proof. The target network update with soft update parameter τ is $Q_{\psi_{\text{target}}}^{k+1} = \tau Q_\psi^k + (1 - \tau) Q_{\psi_{\text{target}}}^k$, $0 < \tau \ll 1$. Then,

$$\|Q_{\psi_{\text{target}}}^{k+1} - Q_{\psi_{\text{target}}}^k\| = \|\tau Q_\psi^k + (1 - \tau) Q_{\psi_{\text{target}}}^k - Q_{\psi_{\text{target}}}^k\| = \tau \|Q_\psi^k - Q_{\psi_{\text{target}}}^k\|. \quad \square$$

Theorem 12 (Transfer Performance Bound). *The performance degradation is estimated by the following expression*

$$|W(\rho_\pi^{\text{real}}, \rho_E) - W(\rho_\pi^{\text{sim}}, \rho_E)| \leq \frac{2L_f L_\pi}{(1 - \gamma)^2} \cdot \mathbb{E}[TV(P_{\text{real}}, P_{\text{sim}})] \quad (44)$$

where $TV(P_{\text{real}}, P_{\text{sim}})(s, a) = \frac{1}{2} \sum_{s' \in S} |P_{\text{real}}(s'|s, a) - P_{\text{sim}}(s'|s, a)|$ represents the expected dynamics model error.

Proof. By using the triangle inequality for Wasserstein distance: $|W(\rho_\pi^{\text{real}}, \rho_E) - W(\rho_\pi^{\text{sim}}, \rho_E)| \leq W(\rho_\pi^{\text{real}}, \rho_\pi^{\text{sim}})$ and

$$\rho_\pi^{\text{real}} = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \rho_\pi^{\text{real}, t}, \quad \rho_\pi^{\text{sim}} = (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t \rho_\pi^{\text{sim}, t}$$

is the discounted state distributions where ρ_π^t is the state distribution at time t . By the convexity of Wasserstein distance

$$W(\rho_\pi^{\text{real}}, \rho_\pi^{\text{sim}}) \leq (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t W(\rho_\pi^{\text{real}, t}, \rho_\pi^{\text{sim}, t}).$$

For any state s , it can be seen from the above facts that the next-state distribution difference is bounded by the inequality

$$W(P_{\text{real}}(\cdot | s, \pi(s)), P_{\text{sim}}(\cdot | s, \pi(s))) \leq L_\pi \cdot \text{TV}(P_{\text{real}}, P_{\text{sim}})$$

where L_π is the Lipschitz constant of the policy. The distribution divergence evolves with the following upper-bound

$$W(\rho_\pi^{\text{real}, t+1}, \rho_\pi^{\text{sim}, t+1}) \leq L_\pi \cdot \mathbb{E}[\text{TV}(P_{\text{real}}, P_{\text{sim}})] + \gamma W(\rho_\pi^{\text{real}, t}, \rho_\pi^{\text{sim}, t}).$$

Solving the recurrence relation $W(\rho_\pi^{\text{real}, t}, \rho_\pi^{\text{sim}, t}) \leq L_\pi \cdot \mathbb{E}[\text{TV}(P_{\text{real}}, P_{\text{sim}})] \cdot \frac{1 - \gamma^t}{1 - \gamma}$, and substituting it gives the equation

$$W(\rho_\pi^{\text{real}}, \rho_\pi^{\text{sim}}) \leq \frac{L_\pi}{1 - \gamma} \cdot \mathbb{E}[\text{TV}(P_{\text{real}}, P_{\text{sim}})] \cdot (1 - \gamma) \sum_{t=0}^{\infty} \gamma^t (1 - \gamma^t) = \frac{L_\pi \gamma}{(1 - \gamma)^2 (1 + \gamma)} \cdot \mathbb{E}[\text{TV}(P_{\text{real}}, P_{\text{sim}})].$$

By absorbing constants and incorporating the Lipschitz constant L_f of the cost function, it can be seen that

$$|W(\rho_\pi^{\text{real}}, \rho_E) - W(\rho_\pi^{\text{sim}}, \rho_E)| \leq \frac{2L_f L_\pi}{(1 - \gamma)^2} \cdot \mathbb{E}[\text{TV}(P_{\text{real}}, P_{\text{sim}})]. \quad \square$$

Theorem 13 (Monotonic Policy Improvement). *Each policy update ensures consistent performance enhancement:*

$$W(\rho_{\pi_k}, \rho_E) - W(\rho_{\pi_{k+1}}, \rho_E) \geq \mathbb{E}[A_{OT}(s, \pi_{k+1}(s))] - 2L_f \cdot W_1(\rho_{\pi_k}, \rho_{\pi_{k+1}}) \quad (45)$$

where $A_{OT}(s, a) = f_k^*(s, a) - \mathbb{E}_{a' \sim \pi_k(\cdot | s)}[f_k^*(s, a')]$ is the OT-based advantage function, avoiding oscillatory behavior.

Proof. We analyze the change in Wasserstein distance between consecutive policies, denoted by the following form

$$\Delta W = W(\rho_{\pi_k}, \rho_E) - W(\rho_{\pi_{k+1}}, \rho_E).$$

By applying the Kantorovich duality for 1-Wasserstein distance $W(\rho_\pi, \rho_E) = \sup_{\|f\|_L \leq 1} [\mathbb{E}_{\rho_E}[f(s)] - \mathbb{E}_{\rho_\pi}[f(s)]]$. Let f_k^* be the optimal Lipschitz function achieving this supremum for policy π_k . By the optimality of f_{k+1}^* for π_{k+1} :

$$W(\rho_{\pi_{k+1}}, \rho_E) \geq \mathbb{E}_{\rho_E}[f_k^*] - \mathbb{E}_{\rho_{\pi_{k+1}}}[f_k^*].$$

Therefore, we obtain the key inequality: $\Delta W \geq \mathbb{E}_{\rho_{\pi_{k+1}}}[f_k^*] - \mathbb{E}_{\rho_{\pi_k}}[f_k^*]$. Considering the Lipschitz property of f_k^* and the definition of Wasserstein distance, it can be directly derived that $|\mathbb{E}_{\rho_{\pi_{k+1}}}[f_k^*] - \mathbb{E}_{\rho_{\pi_k}}[f_k^*]| \leq L_f \cdot W(\rho_{\pi_k}, \rho_{\pi_{k+1}})$. Consider that A_{OT} also satisfies the Lipschitz property $|A_{OT}(s, \pi_{k+1}(s)) - A_{OT}(s', \pi_{k+1}(s'))| \leq L_f \cdot \|s - s'\|$, then

$$\begin{aligned} \left| \mathbb{E}_{s \sim \rho_{\pi_k}}[A_{OT}(s, \pi_{k+1}(s))] - \mathbb{E}_{s \sim \rho_{\pi_{k+1}}}[A_{OT}(s, \pi_{k+1}(s))] \right| &\leq L_f \cdot W_1(\rho_{\pi_k}, \rho_{\pi_{k+1}}), \\ \left| \mathbb{E}_{\rho_{\pi_{k+1}}}[f_k^*] - \mathbb{E}_{\rho_{\pi_k}}[f_k^*] - \mathbb{E}_{s \sim \rho_{\pi_k}}[A_{OT}(s, \pi_{k+1}(s))] \right| &\leq 2L_f \cdot W_1(\rho_{\pi_k}, \rho_{\pi_{k+1}}). \end{aligned}$$

Combining all inequalities gives $W_1(\rho_{\pi_k}, \rho_E) - W_1(\rho_{\pi_{k+1}}, \rho_E) \geq \mathbb{E}[A_{OT}(s, \pi_{k+1}(s))] - 2L_f W_1(\rho_{\pi_k}, \rho_{\pi_{k+1}})$. $\square$

Theorem 14 (End-to-End Convergence). *The complete DOT-PG framework with all five coordinated phases converges to an ϵ -optimal policy:*

$$\limsup_{k \rightarrow \infty} W(\rho_{\pi_k}, \rho_E) \leq \epsilon_{\text{dual}} + \epsilon_{\text{policy}} + \epsilon_{\text{value}} + \epsilon_{\text{target}} \quad (46)$$

where each ϵ term represents the approximation error from the corresponding algorithmic component.

Proof. Let us define $\hat{W}^k$ as the estimated Wasserstein distance using current dual network f_{ϕ_k} , $\hat{W}^{k,*}$ as the optimal Wasserstein distance achievable with perfect dual for policy π_k , and W^* as the global minimum Wasserstein distance over all possible policies. Then, we analyze the total error by decomposing it into several parts as follows

$$W(\rho_{\pi_k}, \rho_E) = \underbrace{W(\rho_{\pi_k}, \rho_E) - \hat{W}^k}_{\epsilon_{\text{dual}}} + \underbrace{\hat{W}^k - \hat{W}^{k,*}}_{\epsilon_{\text{policy}}} + \underbrace{\hat{W}^{k,*} - W^*}_{\epsilon_{\text{value}}} + \underbrace{W^*}_{\epsilon_{\text{target}}}.$$

Note that ϵ_{dual} is the bound of the Dual Network Approximation Error which rises from imperfect estimation of the optimal transport cost and is bounded by Theorem 1 (Maximum Entropy Policy Optimization): $\epsilon_{\text{dual}} \leq \|f_{\phi_k} - f^*\|_{\infty}$, and converges to zero due to alternating optimization (Theorem 5).

Recall that ϵ_{policy} is the bound of the Policy Optimization Error which measures suboptimality of current policy given current dual estimate and is bounded by monotonic improvement guarantee (Theorem 10): $\epsilon_{\text{policy}} \leq \mathbb{E}[A_{OT}(s, \pi_k(s))] - 2L_f W(\rho_{\pi_{k-1}}, \rho_{\pi_k})$, and vanishes due to two-time-scale convergence (Theorem 6).

Notice that ϵ_{value} is the bound of the Value Function Approximation Error which comes from imperfect Q-function estimation and is controlled by target network stabilization (Theorem 7): $\epsilon_{\text{value}} \leq \frac{C}{1-\gamma(1-\tau)} + O(\tau^{-1})$.

Recall that ϵ_{target} is the bound of the irreducible Approximation Error which represents fundamental limitations of policy class and function approximation, and cannot be eliminated but can be minimized through architecture design.

In the proposed coordinated phases, the convergence mechanism is introduced by: 1. Dual Network Phase: $\epsilon_{\text{dual}} \rightarrow 0$ through consistent estimation; 2. Policy Improvement Phase: $\epsilon_{\text{policy}} \rightarrow 0$ via monotonic updates; 3. Value Learning Phase: $\epsilon_{\text{value}} \rightarrow 0$ with temporal difference learning; 4. Target Network Phase: Stabilizes all components; 5. Transfer Learning Phase: Maintains robustness to distribution shift.

By the triangle inequality and individual convergence, $\limsup_{k \rightarrow \infty} W(\rho_{\pi_k}, \rho_E) \leq \epsilon_{\text{dual}} + \epsilon_{\text{policy}} + \epsilon_{\text{value}} + \epsilon_{\text{target}}$. Each term can be made arbitrarily small with sufficient network capacity, training iterations, and suitable hyperparameters. $\square$

Theorem 15 (Sample Complexity). *For $\epsilon > 0$, DOT-PG requires at most:*

$$N(\epsilon) = O\left(\frac{L_f^2 L_{\pi}^2}{(1-\gamma)^4 \epsilon^2} \cdot \log \frac{1}{\delta}\right) \quad (47)$$

samples to achieve $W_1(\rho_{\pi}, \rho_E) \leq \epsilon$ with probability at least $1 - \delta$.

Proof. The sample complexity bound is established through systematic error analysis. The proof considers four primary error sources, each with distinct characteristics and theoretical foundations supported by established literature Bayraktar, E., Eckstein, S., & Zhang, X. (2025), and Fournier, N., & Guillin, A. (2015).

Error Source Analysis: dual network estimation error when estimates the optimal transport cost function f^* scales as $\mathbb{E}[\|f_{\phi} - f^*\|_{\infty}] \leq O(L_f / \sqrt{n_{\text{dual}}})$ which are mainly based on the statistical learning theory for Lipschitz function classes in Wainwright, M. J. (2019). During the optimization, the policy gradient estimator has the variance which is bounded by the inequality $\text{Var}[k] \leq O(L_{\pi}^2 / (1-\gamma)^2 n_{\text{policy}})$ due to cumulative return variance over horizon $1/(1-\gamma)$ Agarwal, A., Kakade, S. M., Lee, J. D., & Mahajan, G. (2021). Q-function approximation error obeys $\mathbb{E}[\|Q_{\psi} - Q^*\|] \leq O(1/\sqrt{n_{\text{value}}})$, while temporal difference amplification yields $\epsilon_{\text{TD}} \leq O(1/(1-\gamma)\sqrt{n_{\text{TD}}})$ from bootstrapping in value estimation, according to Sutton, R. S., & Barto, A. G. (1998).

Lipschitz and Discount Dependencies: The product $L_f L_{\pi}$ emerges from error propagation: policy changes affect state distributions ($\|\rho_{\pi} - \rho_{\pi'}\| \leq L_{\pi} \|\pi - \pi'\|$) which propagate through transport costs ($|f(\rho) - f(\rho')| \leq L_f \|\rho - \rho'\|$), yielding $|W_1(\rho_{\pi}, \rho_E) - W_1(\rho_{\pi'}, \rho_E)| \leq L_f L_{\pi} \cdot \mathbb{E}[\|\pi - \pi'\|]$ via the performance difference lemma Kakade, S., & Langford, J. (2002). The $(1-\gamma)^{-4}$ dependence aggregates four factors: policy gradient variance $O((1-\gamma)^{-2})$ from horizon summation, TD error $O((1-\gamma)^{-1})$ from Bellman contraction, distribution shift accumulation $O((1-\gamma)^{-1})$ from $W_1(\rho_{\pi}^t, \rho_{\pi'}^t) \leq L_{\pi} \sum_{k=0}^{t-1} \gamma^k \|\pi - \pi'\|$, and their worst-case composition.

High-Probability Guarantee: Applying Hoeffding's inequality and union bound over error events E_i : $\mathbb{P}(\bigcup_{i=1}^4 E_i) \leq 4 \exp\left(-c \cdot n \cdot \epsilon^2 / (L_f^2 L_{\pi}^2 (1-\gamma)^{-4})\right)$. Setting this $\leq \delta$ and solving yields $n \geq \frac{CL_f^2 L_{\pi}^2}{(1-\gamma)^4 \epsilon^2} \cdot \log(4/\delta)$. Policy optimization dominates complexity due to nested loops, on-policy sampling, and high-dimensional optimization, giving the final bound $N(\epsilon, \delta) = O\left(\frac{L_f^2 L_{\pi}^2}{(1-\gamma)^4 \epsilon^2} \cdot \log \frac{1}{\delta}\right)$.

The bound justifies Lipschitz regularization for sample efficiency and highlights exponential horizon dependence. For stochastic environments, an additional $\log |\mathcal{S}| |\mathcal{A}|$ term accounts for state-action space size. This analysis completes the

sample complexity proof while demonstrating alignment with established OT and RL theory Genevay, A., Peyré, G., & Cuturi, M. (2018, March) and Peyré, G., & Cuturi, M. (2019). □

7 Conclusion

Your conclusion here

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